ADVANCED PROGRAMMING

Comparative Evaluation of LLM Metrics

Principles, Strengths, Limitations & Applications

Presented by Group 10

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Why Evaluate LLM Outputs?

Different tasks require different definitions of a “good” answer.

THE EVALUATION PROBLEM

Two responses may be equally correct but use very different wording.

EXAMPLE · QUESTION ANSWERING

Reference: Paris

Prediction: The capital is Paris.

01 Correctness

Is the answer correct?

02 Similarity

How closely does it match a reference?

03 Reliability

Can the metric detect meaning changes?

Research question: Which evaluation metric should we use for which NLP task—and why?

Four Perspectives on Evaluation

A practical taxonomy for comparing the nine metric topics.

01 Exact / label-based correctness

EM · Accuracy · Precision / Recall / F1

Strict matches and decisions against ground truth.

02 Lexical similarity

BLEU · ROUGE · METEOR

Shared words, n-grams, sequences and alignments.

03 Semantic similarity

BERTScore

Contextual token-embedding similarities.

04 Edit-based errors

WER · CER

Minimum word or character insertions, deletions and substitutions.

Note: Precision, Recall and F1 are treated as one assigned topic, but they are separate scores.

Exact Match vs. Accuracy

Similar counting formulas, but different correctness rules.

EXACT MATCH (EM)

Is the predicted answer identical to its reference?

EM = exact matches / N

ACCURACY

What fraction of predictions is classified as correct?

Accuracy = correct / N

ILLUSTRATIVE QUESTION

Reference: Paris

Prediction A: Paris → EM = 1

Prediction B: The capital is Paris → EM = 0

Takeaway · EM is string-based; accuracy relies on the task's definition of a correct prediction.

Precision, Recall & F1

Complementary views of classification errors.

CONFUSION MATRIX · EXAMPLE

	PREDICTED +	PREDICTED −
	40 TP	20 FN
	10 FP	30 TN

THE FORMULAS

Precision $P = TP / (TP + FP)$

40 / 50 = 0.800

Recall $R = TP / (TP + FN)$

40 / 60 = 0.667

F1-score $F1 = 2PR / (P + R)$

0.727

Takeaway · F1 balances precision and recall; it is often informative for imbalanced classification.

BLEU vs. ROUGE

Two influential approaches to reference-based lexical overlap.

BLEU

Machine translation

Modified n-gram precision

$$\text{BLEU} = \text{BP} \times \exp\left(\Sigma\, w_n \log p_n\right)$$

- Counts matching candidate n-grams (clipped).
- Adds a brevity penalty for short outputs.
- Traditionally reported at corpus level.

ROUGE

Text summarization

Reference coverage / overlap

$$\text{ROUGE-N} = \text{matched} / \text{ref. n-grams}$$

- ROUGE-1: unigram overlap.
- ROUGE-2: bigram overlap.
- ROUGE-L: longest common subsequence.

Shared limitation · Strong word overlap does not necessarily imply semantic or factual correctness.

Papineni et al. (2002); Lin (2004)

METEOR vs. BERTScore

Linguistic alignment compared with contextual semantic similarity.

DIMENSION	METEOR	BERTSCORE
Underlying idea	Align words and matching chunks	Match contextual token embeddings
Matching signal	Exact forms, stems, synonyms*	Cosine similarity between vectors
Order sensitivity	Explicit fragmentation penalty	Indirectly via contextual embeddings
Typical strength	Handles some lexical variation	Handles semantically related phrasing
Main limitation	Depends on language resources	Needs encoder; not a factuality check

Example: **“The car is fast” ≈ “The automobile is quick”**

**Depending on the METEOR language resources and version.*

Banerjee & Lavie (2005); Zhang et al. (2020)

Word Error Rate vs. Character Error Rate

Both count insertions, deletions and substitutions; their units differ.

A SINGLE MISSING CHARACTER

REFERENCE	h	e	l	l	o
PREDICTION	h	e	l	—	o

1 deletion

WORD ERROR RATE

100%

WER = (S + D + I) / Nw

1 incorrect word / 1 reference word

CHARACTER ERROR RATE

20%

CER = (S + D + I) / Nc

1 deleted character / 5 reference characters

Interpretation · Lower WER and CER are better; scores may exceed 100% with many insertions.

Overall Comparison of the Nine Metric Topics

Every metric captures a different signal and has a distinct blind spot.

METRIC	CORE SIGNAL	COMMON USE	MAIN BLIND SPOT
Exact Match	Whole-answer string	Extractive QA	Rejects paraphrases
Accuracy	Correct labels	Classification	Can hide minority failures
Precision / Recall / F1	Error trade-offs	Classification / QA*	Averaging and definition matter
BLEU	Modified n-grams	Translation	Limited paraphrase recognition
ROUGE	N-grams / subsequence	Summarization	No factuality guarantee
METEOR	Linguistic alignment	Translation	Depends on language resources
BERTScore	Contextual embeddings	Open-text generation	May miss contradictions
WER	Word edit distance	Speech transcription	Depends on segmentation
CER	Character edit distance	OCR / transcription	Ignores meaning

Which Metric Fits Which NLP Task?

Selection should follow the output type and the quality dimension of interest.

Extractive QA	EM + token F1
Text classification	Accuracy + precision / recall / F1
Machine translation	BLEU + METEOR / BERTScore
Text summarization	ROUGE + BERTScore + factuality
Speech transcription	WER + CER
Optical character recognition	CER + WER

THREE FINDINGS

- No universally best metric.
- Similarity is not factuality.
- Use complementary metrics.

Conclusion: metric choice depends on task, reference quality, language, and evaluation objective.

References & Discussion

Foundational works supporting the theoretical comparison.

01	**Papineni, K., et al. (2002).**	BLEU: a Method for Automatic Evaluation of Machine Translation. ACL. https://aclanthology.org/P02-1040/
02	**Lin, C.-Y. (2004).**	ROUGE: A Package for Automatic Evaluation of Summaries. https://aclanthology.org/W04-1013/
03	**Banerjee, S., & Lavie, A. (2005).**	METEOR: An Automatic Metric for MT Evaluation. https://aclanthology.org/W05-0909/
04	**Zhang, T., et al. (2020).**	BERTScore: Evaluating Text Generation with BERT. ICLR. https://arxiv.org/abs/1904.09675
05	**Rajpurkar, P., et al. (2016).**	SQuAD: 100,000+ Questions for Machine Comprehension of Text. https://arxiv.org/abs/1606.05250
06	**JiWER Documentation.**	Word Error Rate and Character Error Rate. https://jitsi.github.io/jiwer/

THANK YOU · QUESTIONS & DISCUSSION